Why the Repair Streams Native AMR Cells

The one-off MPI+Kokkos reducer, without the abstraction tax

GOTHAM PDF reconstruction notes

June 2026

Status: Preferred calculation for additive PDFs. The original repair path is strongly supported; the science catalog and complete Frontier run remain unproven.

1 What we are trying to explain

The task is to rebuild corrupted multidimensional PDFs from full-volume `hydro_w` AMR shards for five simulations and 161 snapshots. The central design question is whether to construct a global dense cube first or to reduce the native AMR cells directly.

The desired object is a weighted histogram. It is not a picture of a uniform grid. That matters.

2 The simple picture

Read one bounded chunk of MeshBlocks. Compute the needed coordinates and weights on the GPU. Add them into rank-local histograms. Throw the chunk away. At the end, sum the histograms across MPI ranks.

The full snapshot never needs to coexist in memory. A dense Cartesian intermediate would be enormous and would introduce interpolation choices that can move mass or signed flux between bins.

3 The cartoon version

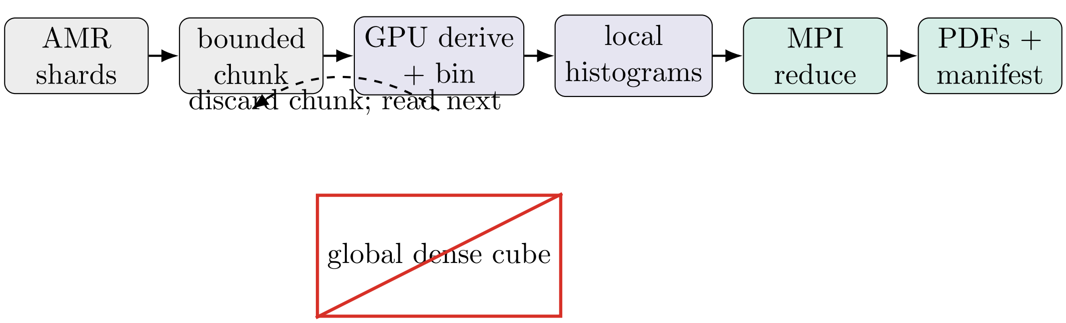

Figure 1: Each bounded AMR chunk contributes to persistent histograms and is then discarded. At no point is a global dense cube assembled. *This is a schematic, not evidence.*

Follow the blue path left to right. The histograms survive each loop; the cell chunk does not.

4 The more precise version

For product p, bin b, and AMR leaf cell c,

$$H_p(b) = \sum_{c\in\text{leaves}} \Delta V_c\, w_p(q_c)\, \mathbf{1}[B_p(x_c, q_c) = b]\,.$$

Here w_p is the pre-volume field: 1 for volume, density for mass, or a flux density for transport products. The stored histogram is an integrated weighted count, not a normalized probability density. Each MPI rank accumulates $H_p^{(r)}$, then

$$H_p = \sum_r H_p^{(r)}.$$

No global cell ordering is needed. The only non-negotiable condition is that the accepted leaf set covers the domain exactly once:

$$\sum_{c\in\text{leaves}} \Delta V_c = V_{\text{domain}},$$

with no duplicate leaf and no ancestor counted alongside a descendant. That is the mathematical requirement. The current implementation checks contiguous shard identity, duplicate logical keys, ancestor/descendant pairs, and summed volume. Those are strong necessary invariants, but they do not prove arbitrary physical coverage: a hole and a distinct-key equal-volume overlap could in principle compensate.

The useful memory claim is about field data, not every byte in the program. A phase-wise peak estimate is

$$M_{\text{peak}} \simeq \max(M_{\text{process}}, M_{\text{validate}}, M_{\text{reduce}}),$$

$$\begin{aligned}
M_{\text{process}} &\simeq M_{\text{host chunk}} + M_{\text{device chunk}} + M_{\text{hist}} + O(N_{\text{blocks},r}),\\
M_{\text{validate,root}} &= O(N_{\text{blocks,global}}),\\
M_{\text{reduce,root}} &\simeq 2M_{\text{hist}} + \max_p(8N_{\text{bins},p}) + O(N_{\text{blocks,global}}).
\end{aligned}$$

There is one line worth noticing: the large field-data term follows the chunk size, not the size of the full snapshot. Metadata and root reduction storage still grow and must be measured at complete-snapshot scale.

One FP64 histogram uses 0.292 GiB for the original repair catalog and 0.325 GiB for all 76 products. Those are not peak-memory measurements. The dominant field-data memory follows `--chunk-blocks`, not the full snapshot. Rank-zero key gathering, product reductions, and output buffers still grow with the complete run and must be measured on Frontier.

Now compare that with the tempting alternative. A uniform 8 pc representation of a 400 kpc domain would contain roughly

$$(400/0.008)^3 = 1.25 \times 10^{14}$$

cells. Eight float32 fields alone would take about 4 PB decimal. That is not merely inconvenient. It is the wrong intermediate for a histogram reduction.

5 What this idea predicts

If direct AMR reduction is really the right calculation, it has to survive six checks.

- A separately implemented cell-by-cell oracle should match every emitted product under the stated physical conventions.

- Changing MPI rank count should not change the scientific result beyond floating-point reduction order.
- Products sharing the same weight should close to the same global total.
- The method should process real shards with bounded field-data memory.
- A deliberately partial run should work but must refuse to claim global AMR closure.
- A complete run should reject missing shards, duplicate logical keys, ancestor/descendant pairs, and wrong summed volume. Stronger geometry checks are needed to reject arbitrary physical holes or overlaps.

Here "complete" means that the job supplies the expected contiguous shard count explicitly. The executable does not infer completeness merely because no input limit was requested.

6 What the evidence actually shows

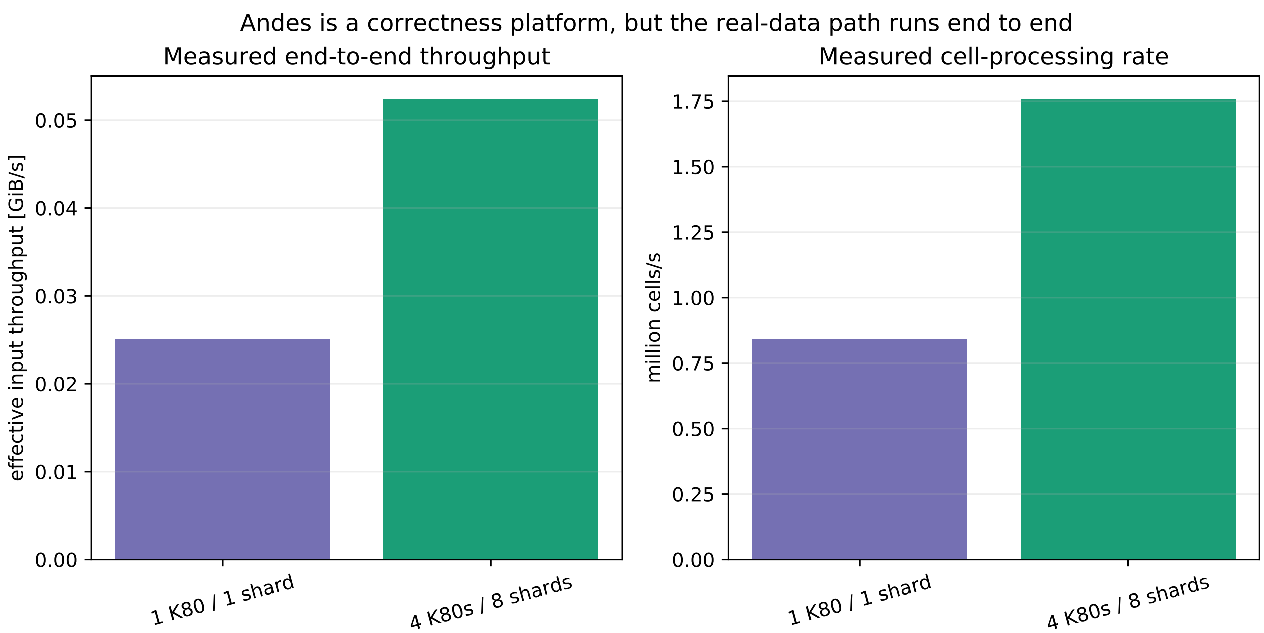

Figure 2: Measured end-to-end rates for one complete real shard on one K80 and eight complete real shards on four K80 GPUs, both using the 14-product `original` catalog. Exact logs are recorded in the evidence audit.

What to look at. Look at the move from one K80 to four K80s. The job processes eight times as much real data, but aggregate throughput improves by only about 2.09×. The useful result is not that K80 is fast. It is not. The useful result is that the full real-data path runs end to end: fixed-record input, device derivation, FP64 histogram accumulation, MPI reduction, and publication. Four GPUs process 20.000 GiB and 671,088,640 cells successfully.

Caveat. The aggregate four-K80 speedup is only about 2.09× relative to the one-K80 rate. This tells us the real path executes, not that production will be efficient. MI250X atomics, I/O overlap, root-memory growth, and GPU-aware reduction still need profiling.

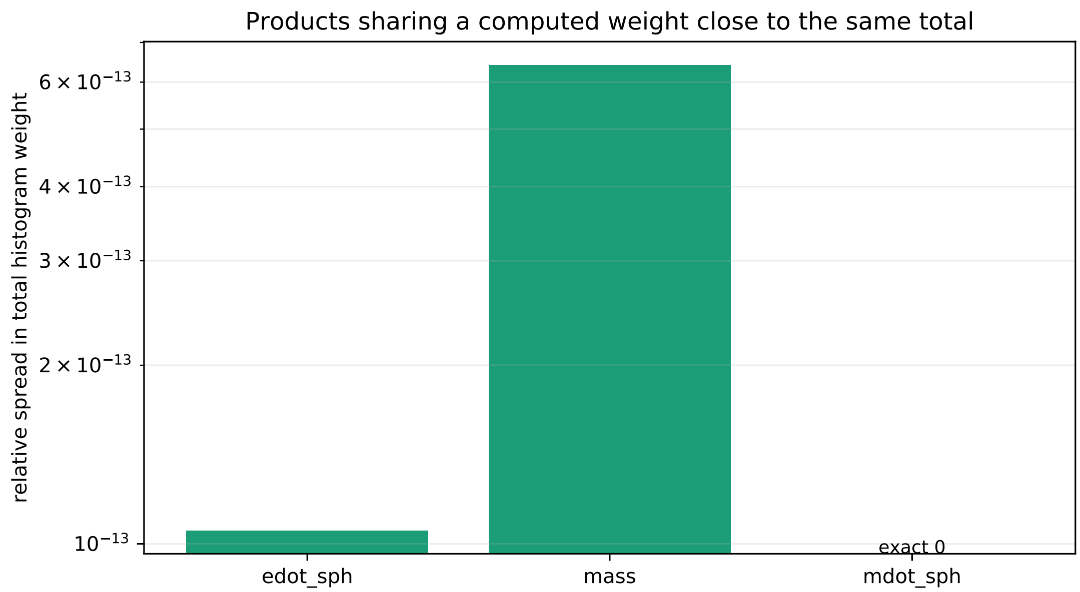

Figure 3: Eight-shard products that use the same weight close to the same total. This is a narrow deposit/reduction invariant, not an external validation of the weight formula.

What to look at. Look at how tightly the same-weight families cluster near zero spread. Their relative differences are around 10^{-13}. Different axis choices redistribute the same computed weight between bins without dropping deposits. A shared wrong weight formula would still pass this test.

7 The key figures

These figures do different jobs. One says the pipeline runs. One says the bookkeeping closes. The corruption document's archived controls ask the harder question: did we reconstruct the right answer?

8 What works

The evidence comes in layers. The separate Python implementation checks the agreed formulas. The one-rank/two-rank tests check decomposition. The real-shard runs check the actual I/O, GPU, MPI, and publication path. The archived controls then check the original energy-flux repair path against surviving external evidence.

- The separately implemented Python oracle covers 14/14 original products and 20/62 science products, plus boundary semantics, signed weights, and synthetic 32^3 and 64^3 block layouts.
- Synthetic one-rank and two-rank outputs agree.
- The production analysis reader accepts the dense reconstructed files.
- One block with all 76 products runs on GPU.
- One complete real shard and an eight-shard/four-GPU real run complete.
- The intact archived energy-flux controls agree closely.

- Publication is fail-closed only for consumers that require the final manifest; individual product files are renamed before that manifest.

9 What does not work

The current kernel is simple enough to audit, and simple enough to bottleneck. Every cell loops over selected products and performs global FP64 atomics. Reading, copying, and kernel work are serialized. MPI reduction is host-staged. Those are plausible performance limits on Frontier.

They are not arguments for a dense cube. They are optimization questions inside this reducer.

This reducer also cannot reconstruct quantities absent from `hydro_w`: magnetic diagnostics, particle history, ion fractions, shock history, or event provenance. Cooling products were excluded because the exact cooling, shielding, and potential reconstruction was not independently validated.

The 62 science products are useful candidates, not externally validated truth. Many use mass, volume, scalar, metal, dust, Mach, angular-momentum, or vertical transport conventions that the surviving archived controls do not test.

10 How this could be wrong

The most serious failure would be a complete snapshot that passes current global invariants but has a physical hole compensated by an equal-volume overlap. The current path validates contiguous shard identity, complete headers, unique leaf keys, ancestor/descendant exclusion, and domain-volume closure. It still needs logical-key bounds and geometry-to-key consistency checks before it can claim exact spatial coverage.

Another risk is semantic drift: "improving" legacy temperature, flux, or underflow conventions would produce scientifically different PDFs. The separate oracle pins the agreed definitions down, but it cannot rule out a shared scientific misunderstanding.

11 What would decide the issue

One complete `res_8pc/phase2/00028` Frontier reconstruction is the next test that can still change our mind about scale. Before claiming exact coverage, add a geometry/key consistency check and a malformed-mesh negative test. Before promoting the science catalog, hand-check representative real cells against independently stated physical conventions. Separate `original`, `science`, and `all` runs then decide whether the current atomic design is fast enough or needs targeted optimization.

12 Bottom line

Bottom line. PDFs are additive reductions over AMR leaves. The streaming MPI+Kokkos reducer performs that operation directly, with bounded field-data memory and without inventing a dense grid that the PDF calculation never needed. The streaming architecture and original repair path are strongly supported. Exact spatial-coverage validation, the broader science catalog, and MI250X performance remain open.

A Revision notes

This document was revised after solution-advocate, logic/technical, evidence/figure, red-team, visual-design, human-editor, and reproducibility checks. Pass two separated architecture validation from science-product validation, narrowed the memory claim, recast same-weight closure as an internal checksum, and weakened exact-coverage and publication claims to match the implementation. Review records live in the local `notes/` directory.